flush, if not a little inside the sheet, thus allowing for the stretch of the ferrule when rolled. In doing this, see to it that the coppers are not rolled too heavy, or the fit will be a poor one.

The fit between the tube end and the copper ferrule should be a driving fit. In flue setting the writer always uses the following system: After all copper ferrules are in place, stuff in all the tubes in the front end. When all are in position and with the helper in the smokebox end, provided with a bar, Fig. 5, lift up and steer the end tube in the *top* row, the man in the firebox catching it with the handle of his hammer and guiding it into the

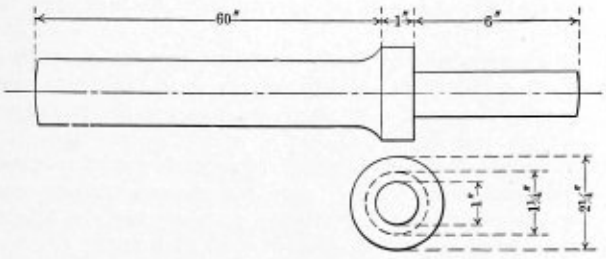


Fig. 5.—Tubing Bar

copper ferrule. Then the helper, with his bar point resting in the tube, gives the tube a few sharp blows on the end, driving it into the copper ferrule. When the required 3/16 inch for bead is through, the man in the firebox knocks the bead, or bell-mouths the tube end, at the same time that the helper is bearing down and drawing back on his bar to prevent it being drawn too far at the firebox end.

Now take the next tube in the *top* row and continue this until the entire set of tubes are knocked over in the firebox. If this is carried out properly, it will be quite impossible to get the tubes crossed. The reason why the writer always knocks the bead all around the tubes (instead of merely knocking a lip on the tubes to hold it in position) is because it is so much *easier* and *quicker* to do it before the tube is fastened in the tube sheet than it is afterwards.

Now take your sectional expander (prosser) and starting, say, in the center of the tube sheet at the bottom, work upwards (right or left) diagonally, taking two rows of tubes at the same time. When this is done you will have formed a V-shape. Then start again at the bottom and work up to the middle of the tube sheet to the top. Starting at either corner, right- or left-hand bottom, work up towards the top center. The reason for doing the work in this manner is to prevent as much as possible the distortion of the tube sheet, which usually takes place in the top corners of the tube sheet and leads to the breaking of the bridge between the tube holes and making the tube holes egg-shape.

USE OF THE SECTIONAL EXPANDER

The writer has recently removed a tube sheet of this kind. In using the sectional expander, great care should be used. It is not wise to drive this tool into a tube either with a maul or a long-stroke air hammer until the mandrel is solid and depend entirely upon the expander to do the work. We are well aware that it is the practice in many of our shops to dispense with the roller-expander in the firebox, but the writer has always used the sectional expander with moderation, knowing the evil resulting from its excessive use; in fact, he has never used it for any other purpose than to *fasten* the tube in the sheet. As a fastening tool it is all right, and, where the tube is a tight fit in the copper ferrule, a slight application of this tool is all that is required. If possible, use the dudgeon rolls for the front end. In beading the tubes, use your own judgment as to doing this before or after rolling; but it has always been the writer's practice to bead after prossering and before rolling. If you have air motors, follow as closely as possible the foregoing.

If every detail explained above is followed out closely, there is very little doubt but that a good and satisfactory job can be accomplished.

Wilkinsburg, Pa. FLEX IBLE.

Leaky Flues

Referring to the letter on leaky flues on page 26 of the January issue of THE BOILER MAKER, in which the writer, Mr. F. Amesbury, of Bridgeburg, Ontario, wished to hear from some of the readers regarding his unfortunate trouble in keeping the flues tight in this particular engine. There is something radically wrong here, and I am very much interested. Flue failures, throughout different railroad roundhouses, have become a very rare thing in these days.

There are several causes of flue failure: First, hard water conditions, which have been reduced to 90 percent by the process of welding the flues after they have been properly inserted; second, the improper setting of copper ferrules and the improper prossering of the flues and leaving them with a large bead; third, flues over 16 feet in length are more apt to give trouble, due to vibration when operating the locomotive; fourth, the practice of welding back tube sheets when bridges are found cracked from one tube hole to the other and increasing the metal to over a half-inch thickness will surely result in failure, and very often this welding will not stand the contraction and expansion of the tube sheet.

WELDING OF FIRE CRACKS

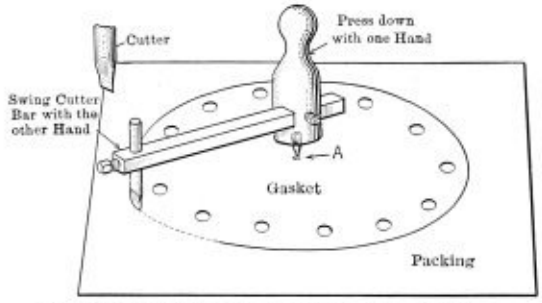
Electric welding of fire cracks and at the rivet holes and building up of wasted flanges in the back tube sheet can be successfully accomplished, and I would suggest that if this engine had been filled up with welds in the flue area and new flues inserted, it would be a good idea, in order to get the mileage out of the flues, to have the flues properly prossered and beaded and the sheet properly cleaned off around the beads and have them electric welded. This will keep the engine in service until the next shopping of locomotive, and then at this time he may apply a new back tube sheet. This will not take this engine out of service over two days. I would also suggest that a brick arch be applied to this engine, if it is not already equipped with one.

It must not be forgotten that a poor fireman and engineer can very easily start a set of flues leaking. From my experience in roundhouse work, I have had locomotives that run as much as a hundred thousand miles showing no leaks in flues at the back tube sheets, and other engines on the same division and of the same type doing the same work, but making only from 50,000 to 60,000 miles. The flues had been worked so that it was necessary to reset the new set of flues, this plainly showing the difference of engineers and firemen. Then it was necessary to shop the engine for a new set of flues. While I was in a hard water district on the Santa Fe in Colorado during 1906, in our passenger engines it was necessary to renew the flues once every 90 days in order to keep the engine in service.

Kearny, N. J. HARRY A. LACERDA,
United States Shipping Board
Emergency Fleet Corporation
Boiler Inspector.

Simple Handy Gasket Cutter

It only takes a few odd minutes and some scrap ends of stock to make this simple device, and it certainly is a handy thing to have in the tool kit. The sketch shows how it is made—just a piece of round stock with a square hole drilled and filed in it to receive the horizontal cutter bar. a set screw to hold it, and a pointed tip center screw, as at



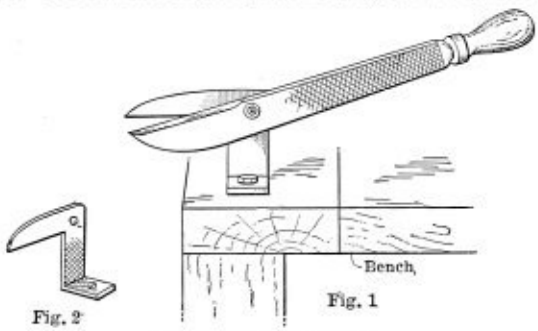
Gasket Cutter Made From Scrap Ends of Stock

A, to pivot the device when cutting the circle. Drill a hole in the end of the square bar to receive the piece of round stock that is used for a cutting knife. Put in a set screw to hold it and the rig is complete. To use it, lay the sheet of packing on a flat board or bench, set the cutter for the largest diameter of the gasket, cut this circle, then set it for the smaller inside circle and cut that.

HELPER.

Bench Shears

There is often a need for small bench shears that enable one man to hold the work in one hand and work the shears with the other. Formerly the way we used to do it was to grip one handle of the tinner's snip in the bench vise,



Improvised Bench Shear

but now we have a pair of improvised bench snips made from two old files. The sketches show how they were made.

Fig. 1 illustrates the lower jaw, which is secured to the bench; Fig. 2, the complete tool. The jaw sections of the file, of course, are ground smooth of file marks.

SHOP MAN.

Staybolts and the Telltale Hole

Some thirty-seven years ago the writer and a friend, now a foreman boiler maker, were walking near the tracks of the old Pittsburgh & Western Railway at Niles, Ohio, and, like all boiler makers, could not pass by a locomotive without looking her over. The day being Sunday, we had lots of time to do so, and the thing that attracted our attention most of all was the staybolts, for they had telltale holes not merely in the ends but clear through the entire length of the bolt. At that time the telltale hole was a novelty in boiler work; that is, they were not in general use and were not applied unless ordered when the engine was built.

Now it is very evident that the telltale holes in these bolts were made right at the mills where the staybolt iron was made and rolled. If I remember rightly, the holes were rather more than 3/16 inch in diameter.

As we all know, the object of the telltale hole is to notify us of the fact that a bolt is fractured and to give us warning of the impending danger. It is a common practice with some engine men to plug up the telltale holes just as soon as they show a leak and say nothing about it when arriving at the end of the division.

On one occasion the writer examined an engine that had been working on another division and found no less than twenty-three broken staybolts in the top row of the side sheet, and every one had the telltale hole plugged, some with the shank of a file driven in and hammered over. Now it would be a very difficult matter for an engineer to do this on the road, for the jacket and lagging would have to be removed to do so; therefore, it is evident that the man on hot work had done the trick to save *himself* a hard job, never giving a thought to the risk the engine crew were running with a boiler in such a condition.

With the staybolts having the telltale hole running through the entire length of the bolt, the matter of plugging up the telltale hole in a broken staybolt becomes a more difficult job for both engine man and boiler maker, for the hostler, when cleaning the fire, would see the leak and report it, even if the engine man had neglected to do so, and the attention of the foreman would be drawn to the fact and he would see that the defective bolts were removed.

For some reason or other it often occurs that the staybolt holes in the wrapper sheet and those in the firebox sheets are not always in line, either leaning to one side or up or down, so that drilling the telltale hole is a matter of guess work and the hole runs to one side of the center of the staybolt. This, of course, applies to drilling after the staybolts are in position. Careful fitting in the first place would avoid all of this trouble, but in new work, when all the holes possible are put into all the sheets, especially the mud-ring holes in back-head, throat, firebox door and flue sheets, before they are flanged (no matter how they are flanged, whether by hand or by press), there is always some variation in the work which will throw the staybolt holes out of line one way or the other and make it almost impossible for anyone to drill in the telltale holes exactly in the center of the bolt as they should be.

By drilling the holes before installation, this trouble would be overcome, but the top-sided staybolt would still remain unless the back-head, throat sheet, firebox door and flue sheets are left blank, so that the mud-ring holes could be put in after fitting. This would allow the fitter-up a chance to adjust the sheets should they be a little out of line in flanging.

The above applies to new work generally, but will also apply to general repairs. It stands to reason that a staybolt, to fulfill its function, should have a straight pull or be as near a right angle to the firebox sheets as possible.

The writer has very recently seen many staybolts give way while the boiler was under test on account of the holes being drilled in after the bolts were in position, the man doing the drilling not knowing that the bolts were out of line. The life of a staybolt out of line is a short one, and it is also a hard bolt to rivet up tight, either by hand or with the machine, for it is never held on solidly.

Allowing that it is impossible always to have our staybolts in perfect line, then the next best thing is to install the bolts with the telltale holes already drilled in. Of course this means that the bolts must be run in from the

on the outside. Where we take the small flues out, we have tools that we use with a small air hammer that we work up from the bottom side of the superheater flues, inside the boiler, and get off all the scale possible.

A Member: How do you rattle them?

A. N. Lucas: In a dry rattler, without any water or pebbles or bolts or anything in them. If you put in the bolts or nuts it smashes your flues up pretty badly. I do not think those large flues ought to be rattled any longer than necessary, and if there is any scale left in spots on it you can knock it off with a hammer.

Mr. Whalen: Our road bought thirty-nine engines three years ago, and we have not removed any of the superheater flues out of those engines yet. When the engines go through the back shop we remove all the two-inch flues and the zig-zag flues up among the superheaters. While they are getting the tubes ready we run a four-inch Lagonda flue cleaner back and forth through each of those superheater flues. You get excellent results from "turbining" those superheater flues. We have never done any tumbling of the superheaters yet.

A. S. Greene: We never clean the superheater flues in the boiler at all; we take them out when they are dirty. We generally take all flues out.

C. R. Bennet: What do you think is the proper way to clean a superheater flue in a boiler?

Mr. Greene: That is something we do not do.

A. N. Lucas: Don't you find that the insides of the superheater flue become heavily coated, and when you take the units out you have hard work getting them out, and if they were not cleaned by the turbine you would have a hard job getting the units in again? We get it a quarter of an inch and five-sixteenths thick where it goes on the inside of the flue.

Mr. Greene: We have not found anything of that kind on our road.

A. N. Lucas: How long do they last before you take them out?

Mr. Greene: Eighteen months to three years.

Mr. Sarver: At our plant we run a set of superheating flues just twice as long as we run the small flues, and, while we have no special way of cleaning those flues other than knocking all the scale off from the outside that we possibly can, in so far as cleaning the scale and stuff off the interior of the flues, I cannot say that we have a whole lot of that to do. If he would say that he has a whole lot of that to do, he is not taking very good care of his superheaters, and probably he is losing a great deal of efficiency on his engines by allowing it to get in that condition.

A. N. Lucas: Have you ever had any units collapse?

Mr. Sarver: We have had them leak on the end, but not had any of the tubes collapsed in the body.

A. N. Lucas: Have you had them pit through at the front flue sheet and fill up the flue from a slow leak?

Mr. Sarver: No, sir.

A. N. Lucas: Have you had any large superheater flues collapse, due to light weight?

Mr. Sarver: Not to my knowledge.

C. R. Bennet: I have had them collapse due to mud, because we were not washing our boilers out right.

Mr. Greene: I have had several superheater flues collapse on the hydrostatic test due to light weight.

Mr. Berry: How do you get the scale out from between the flues after it has accumulated?

A. N. Lucas: Take the flues out.

Mr. Austin: If scale was the result of mud settling and then baking, we might conclude that it was on account of poor boiler washing that scale accumulated; but that is not my understanding of it. Scale accumulates on account of the scale-forming solids in the water being precipitated as the water is evaporated, and it accumulates more where there is the most evaporation, and, like any other cement, it will form regardless of the temperature. We cannot wash it out; it forms there and has got to be knocked off—got to be removed mechanically—you cannot do it with washing.

Mr. Powers: We have recently found it necessary to set a limit to the weight of superheater flues, due to the fact that they collapsed when they were perfectly clean when applied to locomotives in shops. We have adopted a standard of seven pounds to the foot for a 5½-inch flue.

Mr. Tate: My experience in handling superheater tubes is that they wear out at various locations. If that be the case, how are we going to ascertain a defective flue by the weight of it?

A. N. Lucas: Very easily. If it ought to weigh ninety pounds when it is new and you weigh it when it is second-hand and it only weighs seventy, you can decide right quick whether it ought to go in the boiler again or not.

Mr. Tate: That is good, excepting two, four, six or eight inches at the end, where, by piecing that flue out, we have got just as good a flue as when we started.

A. N. Lucas: I doubt it. I have inspected several flues and they did not collapse at the end, but at the center.

Mr. Tate: The only way to ascertain a defective superheater tube is by the hammer test. An experienced boiler maker can find it.

A. N. Lucas: Is that the way you detect a two-inch flue when it is light? Don't you weigh it?

Mr. Tate: Yes.

The Secretary: Mr. Patrick offers the following motion: That when a superheater flue becomes scaled to the extent that it needs cleaning, the best method of cleaning is to remove the flue and clean it in a rattler.

Mr. Wandberg: I have been taking out in the last three or four months superheating flues that have seen six years' service, and the beads on these flues are good, the flues are good.

The question was called for and Mr. Patrick's motion was adopted.

Mr. Austin: I would like to prevent adopting a practice which I think is a mistake, in that it will be putting us to expense and trouble without any return, or with very little return for the trouble and expense. I refer to the practice of using a knocker or rattler on the inside of the large superheater flues to remove the carbon. What little carbon might be in that flue would not amount to much, and it would be practically a waste of labor to go to this expense and trouble to remove it.

Mr. Ready: I find that the cause of the flues collapsing under the hydrostatic test is that in handling the flues in rattling and handling them from the shop to the locomotive they are not as careful as they should be. In lots of cases the flues become pinched or flattened to a certain extent; it may be only for an inch or two, but if you take into consideration the large diameter of the flues and put 175 or 180 pounds water pressure on them, the chances are that you are going to flatten them—going to collapse them. We make it a practice carefully to inspect every superheater flue after it has been cleaned and gotten ready for the boiler, and if we find any kinks or flat places we straighten them with a mandrel that goes into the flue and use an acetylene torch and a small wooden mallet, and have never had a flue collapse since we put this practice in vogue.

roughness remains. You will find the flues have smoothed the holes to a certain extent.

If there is no cutter, the flues will have to be ripped in both ends and knocked back into the back flue sheet until they drop down inside the boiler in the front end, when they can be removed through the big hole. If the boiler is equipped with superheated flues, as most of the large engines of to-day are, and they are to be removed also, it is an easy matter to get the small flues out of the larger holes after the superheater flues are removed.

To remove the superheater flues proceed in the same manner as you would if you were taking out the small flues; that is, after cutting them off inside the front flue sheet, or ripping them and knocking them back, enlarge one of the holes by chipping some of the metal out with a flat chisel and pull the flues through it.

After the flues are all removed, they are placed in a rattler to remove the scale from them, after which a new piece about 6 or 8 inches is welded on to them if they are not too thin or pitted too badly to be used again. They are then cut to the required length and put back into the boiler.

After the flues are removed the burrs or flue ends that remain in the front flue sheet should be removed. All loose dirt and scale should be removed, and the boiler scaled with an air hammer and blunt tools that will not cut into the sheets.

While the flues are in the rattler, and being welded, all rough edges should be removed from the back flue sheet with a rough half-round file and the ferrules applied. A gage should be tried in each hole after the ferrule is applied to see if it is large enough for the flue to enter. This gage can be made of a piece of flue about 8 inches long and swagged to the same size on one end that the flues are to be swagged.

If the flue sheet is warped badly, the gage should fit loosely into the hole, or otherwise the flues which strike some of the holes on an angle, on account of the bulge in the sheet, will be tight, requiring the copper ferrule to be rolled more, which takes considerable time when the flue has to be knocked back and the ferrule rolled. If the sheet is not warped very much, the ferrules need not be rolled so much.

The flues should be a snug fit, but not too tight; they should drive through the ferrules with about 3 or 4 blows from a 10 to 12-pound sledge.

After the ferrules are all finished, the length should be taken, or before they are applied, if the flues are ready to be cut. The length can be taken with a long stick made with a shoulder on one end so it will catch on the outside of the back flue sheet, while a mark is made on the other end of the stick, flush with the outside of the front flue sheet. The stick should not be allowed to sag in the middle or the flues will be too long. It is not necessary to measure every hole, but is safest to measure every other hole, as there is considerable difference between the length, especially when the sheets are warped badly. Beginning with the shortest length, number them 1, 2, 3, etc. Be sure to mark the number over each hole so as not to get the wrong flue in the wrong hole when you are applying them. Also mark the flues.

After the lengths of all the flues are taken, the marks on the stick should be set out a half inch to allow for the bead, which will give you one-quarter inch at both ends, and the flues cut to the required length and annealed on both ends so they can be worked without breaking. The flues are now taken to the engine and are ready to be applied.

Before applying the flues, a copper ferrule should be rolled into the large hole through which the flues were removed. It should be rolled enough so the flue will pass through it.

A man should be put inside the boiler to transfer the flues back of the steam pipes, if they are not removed, and to place them into the back flue sheet.

After the flues are stuck into the boiler they should be set in the firebox end. A large-headed bolt should be placed in the front end of the flues when driving them into the back flue sheet, so as not to damage the flue, which is very soft after being annealed.

If the flue sheet is warped badly, allow a good one-quarter inch for a bead. If it is not warped or cut from the beading tool, a short one-quarter inch is sufficient. If the flues were properly cut to the required length, there will be about one-quarter inch sticking out of the front flue sheet also.

After the flues are set, they should be rolled lightly with a pair of rolls, then expanded with a pair of sectional expanders, driving the pin into the expanders three times.

The flues are next beaded over with a good beading tool that is perfectly smooth and will not leave any ridges or cuts in the flue. Be sure your beading tool is not too large, or it will cut into the sheet.

After the firebox end is finished, cut some strips about 1¼ inches wide, preferably of galvanized iron, to use for shims in the front end. These should be long enough to drive snugly between the flue and sheet, so it will not have to be rolled too much to make it tight, thus weakening it. The shims should be scarfed on both ends so as not to make a lump which will be hard to get tight.

After all the flues are shimmed, give them a good rolling.

After rolling the flues in the front end, some of them can be beaded over, to help brace the flue sheet. The boiler is now ready for the hydrostatic test.

All work should be done with compressed air when it and the tools are available, as it is much quicker than by hand.

Denver, Col. ARTHUR MALET.

Finding the Area of an Irregular Patch

To find the area of an irregular patch: First draw ordinates across the breadth of the patch at equal distances apart, the first and last ordinates each being a half space

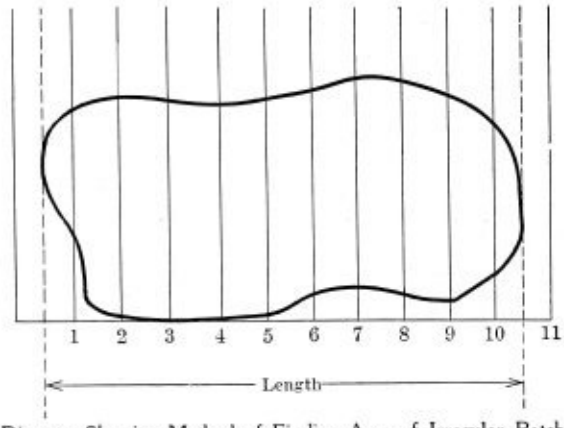


Diagram Showing Method of Finding Area of Irregular Patch

from the ends of the patch. Then find the average breadth by adding together the lengths of these lines included between the boundaries of the patch and dividing by the number of lines added. Multiply this average breadth by the length.

The greatest number of ordinates the more accurate will be the area of the patch.

Oswego, N. Y. JOHN H. HARRISS.

Flues

Methods Employed in Installing Flues in Stationary and Locomotive Boilers

BY GEORGE L. PRICE

In the articles published in THE BOILER MAKER from time to time regarding flues and the different methods employed in the installation of same, I have noticed the differences of opinion as to how to tell when the flue is rolled tight enough by the sound or the drive of the roller pin. According to the drift of this discussion, I am taking it for granted that these writers are talking about rolling flues in stationary boilers, as our present-day methods in locomotive work call for the use of pneumatic tools.

Taking up the question of rolling or expanding flues in stationary boilers first, I will state that when applying flues in watertube boilers that have forged steel headers it is advisable to use precaution when rolling the flues. I am of the opinion that no one but a competent mechanic should attempt this class of work, because watching for the scale to form around the flue on the sheet in this case would not be advisable.

It has been my experience when applying tubes in forged steel headers to take into consideration the ring of the flue pin when being driven into the rolls and the amount of strength required to turn the rolls; also the crease produced by the rolls, the latter, however, to apply only when the ends of the flues are not visible, as when applying the short 6- or 8-inch flues that connect the mud drum. A mechanic may also determine the proper amount of rolling to give a flue by watching the working or expanding process that the end of the flue is undergoing while he is rolling it.

When applying flues on new work, we know that all the flue holes are uniform, and I am of the opinion that the first flue should be rolled by hand. This first flue should be the gage, and, after the flue is rolled in the proper manner, set a stop on your pin and roll all the others in the same manner. This process will insure a uniformity in the flue holes that will be of great aid to the person who puts in the second set, and may possibly save the reaming out of the flue holes when making the second application. And I am sure it will have a tendency to equalize the compression upon the flue sheet bridges, which will give longer life to same.

What cracks our flue sheet bridges? Crystalization. What produces crystalization? The excessive condensation of the molecules in the steel from different causes, as blows of a hammer, variations in temperature, pressure from flue rolls or expanders and other abusive methods which are permitted to exist. Whenever you find a cracked bridge, measure the flue hole adjacent to same and note if it has not been enlarged by the flue rolls or some other agent. I am of the opinion that uniformity should be applied in all cases, or, in other words, we should adopt a standard practice in our work and constantly apply same.

When rolling flues on new work in stationary boilers with steel heads it is often the practice in many shops to have a boiler-maker helper roll the front end while the boiler maker is beading the back end. In this case, I would again advise the use of the stop on the flue pin, said stop, however, to be set by the boiler maker. This method will not permit excessive rolling by the inexperienced man. Flues in all classes of boilers should be shimmed tight in the proper manner before they are rolled. A flue that is loose in the hole when it is being rolled will only thin out and possibly burst in the seam.

I do not think it is good policy for the flue ends to protrude out of the sheet over ½ inch when they are not beaded. The end to be beaded should not protrude out of the sheet more than 3/16 inch, and up to 5/16 inch the length for bead is generally regulated by the style and size of the cavity of the beading tool. Flues in firetube boilers are generally beaded, while flues in watertube boilers are not.

The question is often asked why do we bead the fire end of the tube and not the other? Generally on stationary boilers of the horizontal type both ends are beaded, but on a vertical boiler the flues are generally beaded on the fire end only. They are beaded on the fire end to protect the ends of the flues, as the excessive heat from the hot gases strikes them very forcibly upon entering the flue in its passage to the front end. The wear and tear of this constant force upon an unprotected flue end would soon cause considerable trouble. On the other hand, at the front end of the tube the hot gases, in departing from the tube ends, do not have the same forcible effect on account of their reduced temperature and their travel out of the tubes.

When rolling tubes by air the self-feeding expanders should be used with a stop on the p'n. However, I do not think it is advisable to roll flues from 2½ to 4 inches with air, as this operation necessitates the use of a high-powered machine, which is very heavy to handle and also dangerous. When using the self-feeding rolls with an air machine the stop on the pin will pull the pin automatically when the pin reaches the proper point of contact on the stop. This operation saves the workman the time and trouble of reversing his air machine on every flue.

It is very good practice to have both sides of the flue hole chamfered off slightly when the flues are to be prossered, thereby eliminating the sharp edge of the flue hole, which will have a tendency to shear the flue when it is worked. When the flues are to be expanded with the roller expander the flue hole should be chamfered off on the side of the sheet where the flues are to be beaded. I also believe in a driving fit for all flues wherever it is possible.

LOCOMOTIVE WORK

In locomotive work the flues are generally rolled or expanded with air tools in the front end and not prossered. I consider a flue that protrudes ¼ inch outside of the front flue sheet the proper length and the most convenient to work, for when we get a flue that protrudes from ½ inch to 1 inch outside the sheet, and especially when the holes are worn large, they are liable to split when they are pinned out.

All flues are generally shimmed in the front end with iron shims. The use of copper shims has been abandoned by the railroads for some time, and I do not think copper shims are necessary when sheet iron answers the purpose just as well. On new work, when the sheets are new, some railroads do not use shims in the front end. They enlarge the flue ends at the flue fire so that they fit into